

Book I.

Definitions

1.

A point is that which has no parts.

2.

A line is length without breadth.

3.

The extremities of a line are points.

4.

The straight or right line is that which lies evenly between its extremities.

5.

A surface is that which has length and breadth only.

6.

The extremities of a surface are lines.

7.

A *plane surface* is that which lies evenly between its extremities.

8.

A *plane angle* is the inclination of two lines to one another, in a plane, which meet together, but are not in the same direction.



9.

A *plane rectilinear angle* is the inclination of two straight lines to one another, which meet together, but are not in the same straight line.



10.

When one straight line standing on another straight line makes the adjacent angles equal, each of these angles is called a *right angle*, and each of these lines is said to be *perpendicular* to one another.



11.

An *obtuse angle* is an angle greater than a right angle.



12.

An *acute angle* is an angle less than a right angle.

13.

A *term* or *boundary* is the extremity of any thing.

14.

A *figure* is a surface enclosed on all sides by a line or lines.



15.

A *circle* is a plane figure, bounded by one continued line, called its circumference or periphery; and having a certain point within it, from which all straight lines drawn to its circumference are equal.

16.

This point (from which the equal lines are drawn) is called the *centre* of the circle.



17.

A *diameter* of a circle is a straight line drawn through the centre, terminated both ways in the circumference.



18.

A *semicircle* is the figure contained by the diameter, and the part of the circle cut off by the diameter.



19.

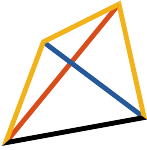
A *segment* of a circle is a figure contained by a straight line and the part of the circumference which it cuts off.

20.

A figure contained by straight lines only, is called a *rectilinear figure*.

21.

A *triangle* is a rectilinear figure included by three sides.



22.

A *quadrilateral figure* is one which is bounded by four sides. The straight lines  and  connecting the vertices of the opposite angles of a quadrilateral figure, are called its *diagonals*.

23.

A *polygon* is a rectilinear figure bounded by more than four sides.



24.

A triangle whose sides are equal, is said to be *equilateral*.



25.

A triangle which has only two sides equal is called an *isosceles triangle*.

26.

A *scalene triangle* is one which has no two sides equal.



27.

A *right angled triangle* is that which has a right angle.



28.

An *obtuse angled triangle* is that which has an obtuse angle.



29.

An *acute angled triangle* is that which has three acute angles.



30.

Of four-sided figures, a *square* is that which has all its sides equal, and all its angles right angles.



31.

A *rhombus* is that which has all its sides equal, but its angles are not right angles.



32.

An *oblong* is that which has all its angles right angles, but has not all its sides equal.



33.

A *rhomboid* is that which has its opposite sides equal to one another, but all its sides are not equal nor its angles right angles.

34.

All other quadrilateral figures are called *trapeziums*.



35.

Parallel straight lines are such as are in the same plane, and which being produced continually in both directions would never meet.

Book I.

Postulates

I.

Let it be granted that a straight line may be drawn from any one point to any other point.

II.

Let it be granted that a finite straight line may be produced to any length in a straight line.

III.

Let it be granted that a circle may be described with any centre at any distance from that centre.

Book I.

Axioms

I.

Magnitudes which are equal to the same are equal to each other.

II.

If equals be added to equals the sums will be equal.

III.

If equals be taken away from equals the remainders will be equal.

IV.

If equals be added to unequals the sums will be unequal.

V.

If equals be taken away from unequals the remainders will be unequal.

VI.

The doubles of the same or equal magnitudes are equal.

VII.

The halves of the same or equal magnitudes are equal.

VIII.

The magnitudes which coincide with one another, or exactly fill the same space, are equal.

IX.

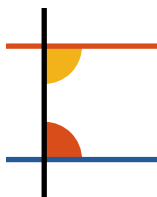
The whole is greater than its part.

X.

Two straight lines cannot include a space.

XI.

All right angles are equal.



XII.

If two straight lines (——— and ———) meet a third straight line (———) so as to make the two interior angles ( and ) on the same side less than two straight angles, these two straight lines will meet if produced on that side on which the angles are less than two right angles.

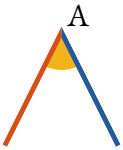
The twelfth axiom may be expressed in any of the following ways:

1. Two diverging straight lines cannot be both parallel to the same straight line.
2. If a straight line intersect one of the two parallel straight lines it must also intersect the other.
3. Only one straight line can be drawn through a given point, parallel to a given straight line.

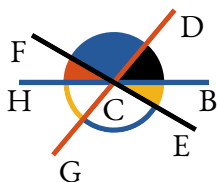
Book I.

Elucidations

Geometry has for its principal objects the exposition and explanation of the properties of *figure*, and figure is defined to be the relation which subsists between the boundaries of space. Space or magnitude is of three kinds, *linear*, *superficial*, and *solid*.



Angles might properly be considered as a fourth species of magnitude. Angular magnitude evidently consists of parts, and must therefore be admitted to be a species of quantity. The student must not suppose that the magnitude of an angle is affected by the length of the straight lines which include it and of whose mutual divergence it is the measure. The *vertex* of an angle is the point where the *sides* or *legs* of the angle meet, as A.



An angle is often designated by a single letter when its legs are the only lines which meet together at its vertex. Thus the red and blue lines form the yellow angle, which in other systems would be called angle A. But when more than two lines meet in the same point, it was necessary by former methods, in order to avoid confusion, to employ three letters to designate an angle about that point, the letter which marked the vertex of the angle being always placed in the middle. Thus the black and red lines meeting together at C, form the blue angle, and has been usually denominated the angle FCD or DCF. The lines FC and CD are the legs of the angle; the point C is its vertex. In like manner the black angle would be designated the angle DCB or BCD. The red and blue angles added together, or the angle HCF added to FCD, make the angle HCD; and so of other angles.

When the legs of an angle are produced or prolonged beyond its vertex, the angles made by them on both sides of the vertex are said to be *vertically opposite* to each other: thus the red and yellow angles are said to be vertically opposite angles.

Superposition is the process by which one magnitude may be conceived to be placed upon another, so as exactly to cover it, or so that every part of each shall exactly coincide.

A line is said to be *produced*, when it is extended, prolonged, or it has its length increased, and the increase of length which it receives is called the *produced part*, or its *production*.

The entire length of the line or lines which enclose a figure, is called its

perimeter. The first six books of Euclid treat of plane figures only. A line drawn from the centre of a circle to its circumference, is called a *radius*. That side of a right angled triangle, which is opposite to the right angle, is called the *hypotenuse*. An oblong is defined in the second book, and called a *rectangle*. All lines which are considered in the first six books of the Elements are supposed to be in the same plane.

The *straight-edge* and *compasses* are the only instruments, the use of which is permitted in Euclid, or plain Geometry. To declare this restriction is the object of the *postulates*.

The *Axioms* of geometry are certain general propositions, the truth of which is taken to be self-evident and incapable of being established by demonstration.

Propositions are those results which are obtained in geometry by a process of reasoning. There are two species of propositions in geometry, *problems* and *theorems*.

A *Problem* is a proposition in which something is proposed to be done; as a line to be drawn under some given conditions, a circle to be described, some figure to be constructed, &c. The *solution* of the problem consists in showing how the thing required may be done by the aid of the straight-edge and compasses. The *demonstration* consists in proving that the process indicated in the solution attains the required end.

A *Theorem* is a proposition in which the truth of some principle is asserted. This principle must be deduced from the axioms and definitions, or other truths previously and independently established. To show this is the object of demonstration.

A *Problem* is analogous to a postulate.

A *Theorem* resembles an axiom.

A *Postulate* is a problem, the solution to which is assumed.

An *Axiom* is a theorem, the truth of which is granted without demonstration.

A *Corollary* is an inference deduced immediately from a proposition.

A *Scholium* is a note or observation on a proposition not containing an inference of sufficient importance to entitle it to the name of *corollary*.

A *Lemma* is a proposition merely introduced for the purpose of establishing some more important proposition.

Book I.

Symbols and Abbreviations

$\therefore$	expresses the word <i>therefore</i> .
$\because$	expresses the word <i>because</i> .
$=$	expresses the word <i>equal</i> . This sign of equality may be read <i>equal to</i> , or <i>is equal to</i> , or <i>are equal to</i> ; but the discrepancy in regard to the introduction of the auxiliary verbs <i>is</i> , <i>are</i> , &c. cannot affect the geometrical rigour.
$\neq$	means the same as if the words ' <i>not equal</i> ' were written.
$>$	signifies <i>greater than</i> .
$<$	signifies <i>less than</i> .
$\nless$	signifies <i>not greater than</i> .
$\nless$	signifies <i>not less than</i> .
$+$	is read <i>plus (more)</i> , the sign of addition; when interposed between two or more magnitudes, signifies their sum.
$-$	is read <i>minus (less)</i> , signifies subtraction; and when placed between two quantities denotes that the latter is taken from the former.

× expresses the product of two or more numbers when placed between them in arithmetic and algebra; but in geometry it is generally used to express a *rectangle*, when placed between “two straight lines which contain one of its right angles.” A *rectangle* may also be represented by placing a point between two of its conterminous sides.

∴ expresses an *analogy* or *proportion*; thus if A, B, C and D represent four magnitudes, and A has to B the same ratio that C has to D, the proportion is written $A : B :: C : D$, $A : B = C : D$, or $\frac{A}{B} = \frac{C}{D}$.

This equality is read: *as A is to B, so is C to D.*

∥ signifies *parallel to*.

⊥ signifies *perpendicular to*.

△ signifies *angle*.

∟ signifies *right angle*.

⌒ signifies *two right angles*.

↑ or briefly designates a *point*.

↯

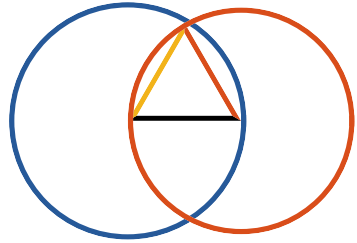
The square described on a line is concisely written thus, —^2 . In the same manner twice the square of, is expressed by $2 \cdot \text{—}^2$.

def. signifies *definition*.

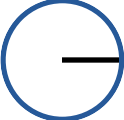
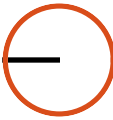
post. signifies *postulate*.

- ax. signifies *axiom*.
- hyp. signifies *hypothesis*. The *hypothesis* is the condition assumed or taken for granted. Thus, the hypothesis of the proposition that a triangle is isosceles is that its legs are equal.
- const. signifies *construction*. The *construction* is the change made in the original figure, by drawing lines, making angles, describing circles, &c., in order to adapt it to the argument of the demonstration or the solution of the problem. The conditions under which these changes are made are as indisputable as those contained in the hypothesis.
- Q. E. D. signifies *Quod erat demonstrandum* — which was to be demonstrated.

Book I. Prop. I. Prob.



*On a given finite straight line
() to describe an equilateral
triangle.*

Describe  and  (post. III);

draw  and  (post. I).

Then will  be equilateral.

For  =  (def. 15);

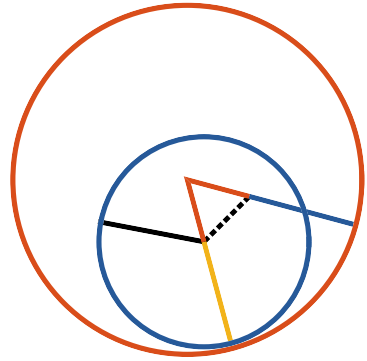
and  =  (def. 15);

$\therefore$  =  (ax. I);

and therefore  is the equilateral triangle required.

Q. E. D.

Book I. Prop. II. Prob.



*From a given point (———), to draw
a straight line equal to a given straight
line (———).*

Draw (post. I), describe  (pr. I.1),

produce ——— (post. II),

describe  and  (post. III);

produce ——— (post. II),

then ——— is the line required.

For ——— = ——— (def. 15),

and ——— = ——— (const.),

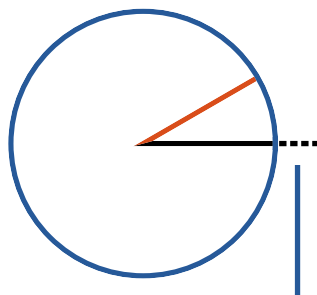
∴ ——— = ——— (post. III),

but (def. 15) ——— = ——— = ——— ;

∴ ——— drawn from the given point (———), is equal to the given
line ——— .

Q. E. D.

Book I. Prop. III. Prob.



From the greater (— — —) of two given straight lines, to cut off a part equal to the less (— — —).

Draw — — — = — — — (pr. I.2);

describe  (post. III),

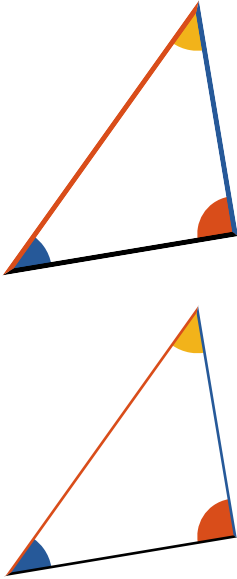
then — — — = — — — .

For — — — = — — — (def. 15),

and — — — = — — — (const.);

∴ — — — = — — — (ax. I).

Q. E. D.



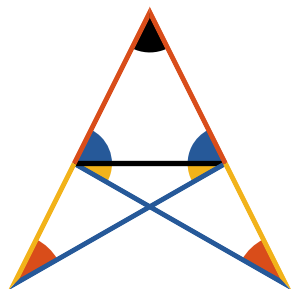
Book I. Prop. IV. Theor.

If two triangles have two sides of the one respectively equal to two sides of the other, (— to — and — to —) and the angles (◡ and ◡) contained by those equal sides also equal; then their bases or their sides (— and —) are also equal: and the remaining angles opposite to equal sides are respectively equal (◡ = ◡ and ◡ = ◡): and the triangles are equal in every respect.

Let two triangles be conceived, to be so placed, that the vertex of one of the equal angles (◡ or ◡) shall fall upon that of the other, and — to coincide with —, then will — coincide with — if applied: consequently — will coincide with —, or two straight lines will enclose a space, which is impossible (ax. X), therefore — = —, ◡ = ◡ and ◡ = ◡,

and as the triangles ◡ and ◡ coincide, when applied, they are equal in every respect.

Q. E. D.



Book I. Prop. V. Theor.

In any isosceles triangle  if the equal sides be produced, the external angles at the base are equal, and the internal angles at the base are also equal.

Produce  and  (post. II),
take  =  (pr. I.3);
draw  and .

Then in  and 

we have  =  (const.),

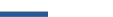
 common to both,

and  =  (hyp.)

$\therefore$  = ,  =  and
 =  (pr. I.4).

Again in  and 

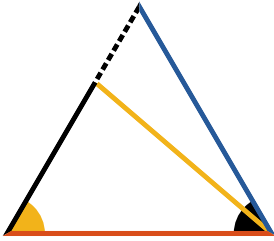
we have  = ,  = 

and  = ,

$\therefore$  =  and  = 
(pr. I.4)

but  = , $\therefore$  = 
(ax. III).

Q. E. D.



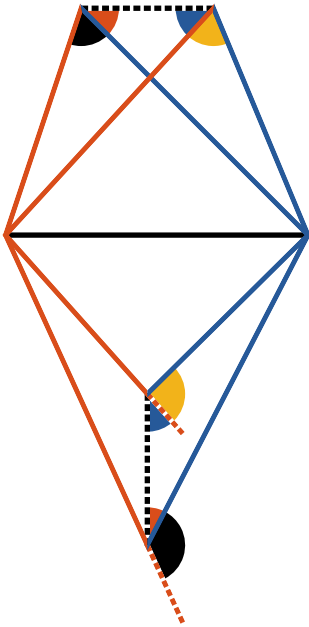
Book I. Prop. VI. Theor.

In any triangle () if two angles ( and ) are equal, the sides ( and ) opposite to them are also equal.

For if the sides be not equal, let one of them  be greater than the other , and from it cut off  =  (pr. I.3), draw .

Then in  and ,
 =  (const.),
 =  (hyp.)
 and  common,
 $\therefore$ the triangles are equal (pr. I.4)
 a part equal to the whole, which is
 absurd;
 $\therefore$ neither of the sides  or
 is greater than the other,
 hence they are equal.

Q. E. D.



Book I. Prop. VII. Theor.

On the same base (—), and on the same side of it there cannot be two triangles having their conterminous sides (— and —, — and —) at both extremities of the base, equal to each other.

When two triangles stand on the same base, and on the same side of it, the vertex of the one shall either fall outside of the other triangle, or within it; or, lastly, on one of its sides.

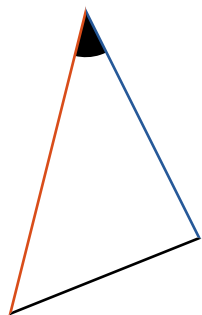
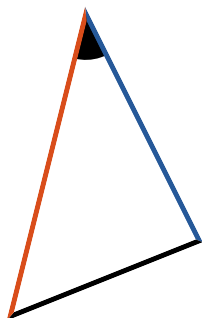
If it be possible let the two triangles be constructed so that

$$\left\{ \begin{array}{l} \text{—} = \text{—} \\ \text{—} = \text{—} \end{array} \right\}, \quad \text{then draw}$$
 and,

$$\left. \begin{array}{l} \blacktriangle = \triangle \text{ (pr. I.5)} \\ \therefore \blacktriangle < \triangle \text{ and} \\ \therefore \blacktriangle < \triangle \end{array} \right\} \text{ which is absurd,}$$
 but (pr. I.5) $\blacktriangle = \triangle$

therefore the two triangles cannot have their conterminous sides equal at both extremities of the base.

Q. E. D.



Book I. Prop. VIII. Theor.

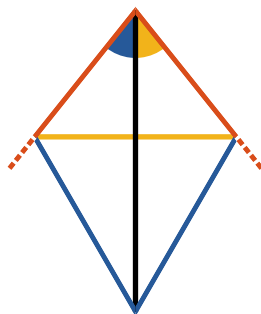
If two triangles have two sides of the one respectively equal to two sides of the other (— = — and — = —) and also their bases (— = —), equal; then the angles (◼ and ◼) contained by their equal sides are also equal.

If the equal bases — and — be conceived to be placed one upon the other, so that the triangles shall lie at the same side of them, and that the equal sides — and —, — and — be conterminous, the vertex of the one must fall on the vertex of the other; for to suppose them not coincident would contradict the last proposition.

Therefore sides — and —, being coincident with — and —, $\therefore \blacktriangle = \blacktriangle$.

Q. E. D.

Book I. Prop. IX. Prob.



To bisect a given rectilinear angle
()

Take  =  (pr. I.3)

draw , upon which describe  (pr. I.1),
draw .

$\therefore$  =  (const.)

and  common to the two triangles

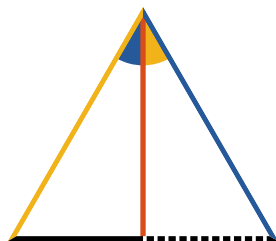
and  =  (const.),

$\therefore$  =  (pr. I.8).

Q. E. D.

Book I. Prop. X. Prob.

To bisect a given finite straight line
(**— ····**).

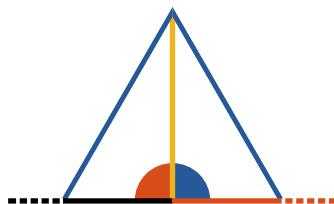


Construct  (pr. I.1),
draw , making  =  (pr. I.9).
Then  =  by (pr. I.4),
for  =  (const.)  = 
and  common to the two triangles.
Therefore the given line is bisected.

Q. E. D.

Book I. Prop. XI. Prob.

From a given point (), in a
given straight line (), to
draw a perpendicular.



Take any point () in the given line,
cut off  =  (pr. I.3),

construct  (pr. I.1),
draw  and it shall be perpendicular to the given line.

For  =  (const.)

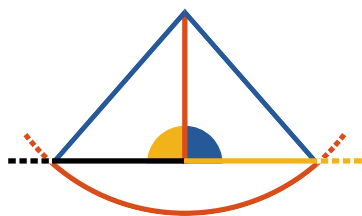
 =  (const.)

and  common to the two triangles.

Therefore  =  (pr. I.8)

$\therefore$  $\perp$  (def. 10).

Q. E. D.



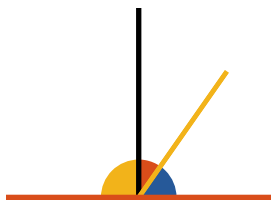
Book I. Prop. XII. Prob. *To draw a straight line perpendicular to a given indefinite straight line (— —) from a given point (↑) without.* With the

given point ↑ as centre, at one side of the line, and any distance — — capable of extending to the other side, describe ⤿.

Make — — = — — (pr. I.10), draw — —, — — and — —, then — — ⊥ — —.

For (pr. I.8) since — — = — — (const.), — — common to both, and — — = — — (def. 15),
 $\therefore$ — — = — —, and
 $\therefore$ — — ⊥ — — (def. 10).

Q. E. D.



Book I. Prop. XIII. Theor. *When a straight line (—) standing upon another straight line (—) makes angles with it; they are either two right angles or together equal to two right angles. If — be $\perp$ to — then,*

 and  =  (def. 10),

but if — be not $\perp$ to —, draw — $\perp$ — (pr. I.11);

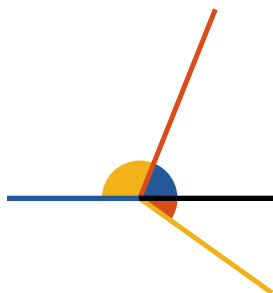
 +  =  (const.),

 =  =  + 

$\therefore$  +  =  +  +  (ax. II)

=  +  = .

Q. E. D.



Book I. Prop. XIV. Theor. *If two straight lines (— and —), meeting a third straight line (—), at the same point, and at opposite sides of it, make with it adjacent angles (and) equal to two right angles; these straight lines lie in one continuous straight line. For, if possible let —,*

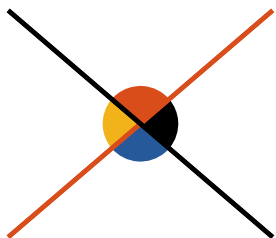
and not —, be the continuation of —, then $\text{yellow arc} + \text{blue arc} = \text{semicircle}$

but by the hypothesis $\text{yellow arc} + \text{blue arc} = \text{semicircle}$

$\therefore \text{blue arc} = \text{blue arc}$ (ax. III); which is absurd (ax. IX).

$\therefore$ — is not the continuation of —, and the like may be demonstrated of any other straight line except —, $\therefore$ — is the continuation of —.

Q. E. D.



Book I. Prop. XV. Theor. *If two right lines (— and —) intersect one another, the vertical angles  and ,  and  are equal.*

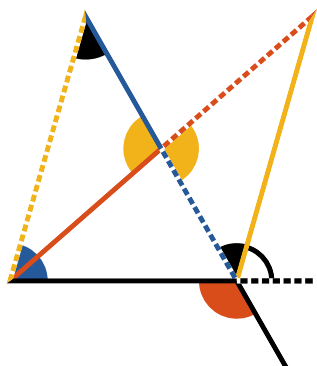


$$\therefore \text{yellow angle} = \text{black angle}.$$

In the same manner it may be shown
that

$$\text{red angle} = \text{blue angle}.$$

Q. E. D.



Book I. Prop. XVI. Theor. *If a*

side of a triangle () is produced, the external angle () is greater than either of the internal remote angles ( or ). Make  =

 (pr. I.10);

draw  and produce it until

 =  ;

draw  .

In  and  ;

 =  ,  =  and

 =  (const., pr. I.15),

$\therefore$  =  (pr. I.4),

$\therefore$  >  .

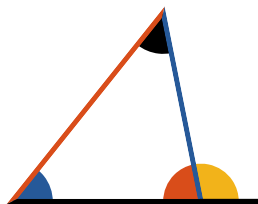
In like manner it can be shown, that if

 be produced,  > 

and therefore  which is = 

is >  .

Q. E. D.



Book I. Prop. XVII. Theor. *Any two*

angles of a triangle  *are together less than two right angles.*

Produce , then will

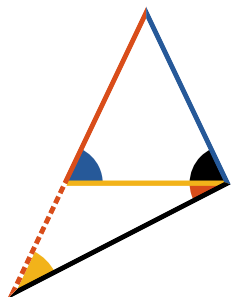
$$\text{red sector} + \text{yellow sector} = \text{semicircle}.$$

But  $>$  (pr. I.16)

$$\therefore \text{red sector} + \text{blue sector} < \text{semicircle},$$

and in the same manner it may be shown that any other two angles of the triangle taken together are less than two right angles.

Q. E. D.



Book I. Prop. XVIII. Theor. *In any*

triangle  *if one side*  *be*
greater than another  *, the an-*
gle opposite to the greater side is greater
than the angle opposite to the less. I.e.

 $>$ .

Make  $=$  (pr. I.3), draw
.

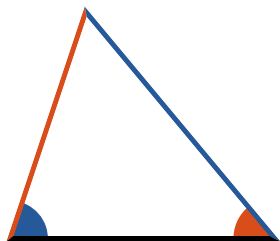
Then will  $=$  (pr. I.5);

but  $>$  (pr. I.16)

$\therefore$  $>$  and much more

is  $>$ .

Q. E. D.



Book I. Prop. XIX. Theor. *If*

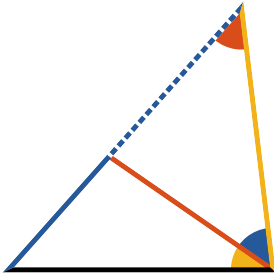
in any triangle  *one angle*  *be greater than another*  *the side*  *which is opposite to the greater angle, is greater than the side*  *opposite the less.*

If  be not greater than 
then must
 = or < .

If  =  then
 =  (pr. I.5);
which is contrary to the hypothesis.

 is not less than ; for if it
were,
 <  (pr. I.18)
which is contrary to the hypothesis:
 $\therefore$  > .

Q. E. D.



Book I. Prop. XX. Theor. *Any two sides  and  of a triangle  taken together are greater than the third side ().*

Produce , and
make  =  (pr. I.3);
draw .

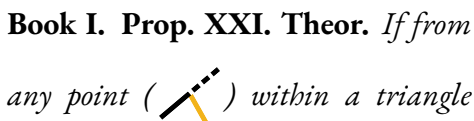
Then $\because$  =  (const.),
 =  (pr. I.5)

$\therefore$  >  (ax. IX)

$\therefore$  +  >
 (pr. I.19)

and $\therefore$  +  > .

Q. E. D.



 , straight lines be drawn to the extremities of one side (-----), these lines must be together less than the other two sides, but must contain a greater angle.

Produce $\overline{\text{---}}$,
 $\text{---} + \text{---} > \overline{\text{---}}$ (pr. I.20),
 add --- to each,
 $\text{---} + \text{---} > \text{---}$
 $\overline{\text{---}} + \text{---}$ (ax. IV)

in the same manner it may be shown
that

$12 + 5 = 17$

which was to be proved.

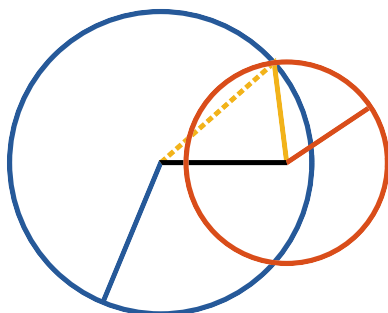
Again $\blacktriangleleft > \blacktriangleright$ (pr. I.16),

and also $\triangle ABC > \triangle DEF$ (pr. I.16),

A diagram illustrating a comparison between two sets. On the left, there is a set of three points (represented by three small black dots) and a large orange triangle. In the center is a greater-than sign (>). On the right is a large yellow triangle and a set of two points (represented by two small black dots).

Q. E. D.

Book I. Prop. XXII. Prob.



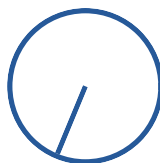
Given three right lines {    the sum of any two greater than the third, to construct a triangle whose sides shall be respectively equal to the given lines.



Assume  =  (pr. I.3).

Draw  =  } (pr. I.2).
and  = 

With  and  as radii, describe



and



(post. III);

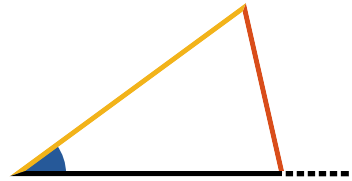
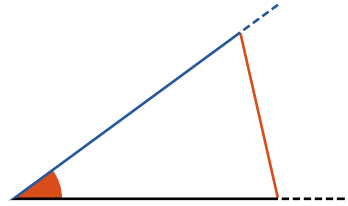
draw  and ,

then will  be the triangle required.

For  = ,
 =  = , } (const.)
and  =  = .

Q. E. D.

Book I. Prop. XXIII. Prob.



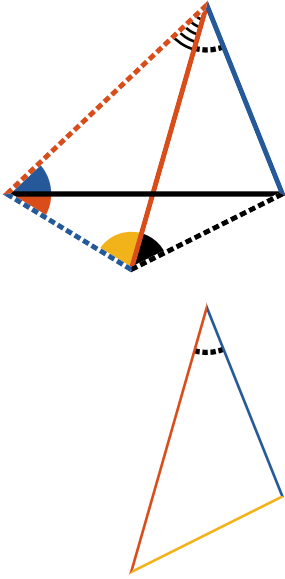
At a given point () in a given straight line (), to make an angle equal to a given rectilinear angle ().

Draw  between any two points in the legs of the given angle.

Construct  (pr. I.22)
 so that  = ,  = 
 and  = .

Then  =  (pr. I.8).

Q. E. D.



Book I. Prop. XXIV. Theor. *If*

two triangles have two sides of the one respectively equal to two sides of the other (— to — and to —), and if one of the angles ($\triangle$) contained by the equal sides be greater than the other ($\triangle$), the side (—) which is opposite to the greater angle is greater than the side (—) which is opposite to the less angle.

Make $\triangle = \triangle$ (pr. I.23),
and — = — (pr. I.3),
draw and

$\therefore$ — =
..... (ax. I, hyp., const.)

$\therefore$ $\triangle = \triangle$ (pr. I.5)

but $\triangle < \triangle$,

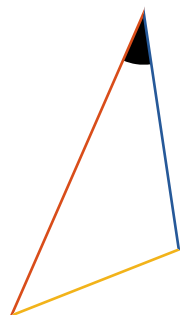
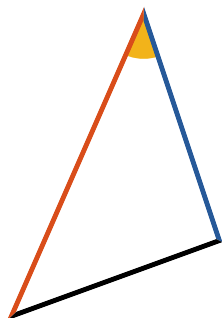
and $\therefore \triangle < \triangle$,

$\therefore$ — > (pr. I.19)

but = — (pr. I.4)

$\therefore$ — > — .

Q. E. D.



Book I. Prop. XXV. Theor. *If two triangles have two sides (— and —) of the one respectively equal to two sides (— and —) of the other, but their bases unequal, the angle subtended by the greater base (—) of the one, must be greater than the angle subtended by the less base (—) of the other.*

▲ = , > or < ▲

▲ is not equal to ▲

for if ▲ = ▲ then

— = — (pr. I.4)

which is contrary to the hypothesis;

▲ is not less than ▲

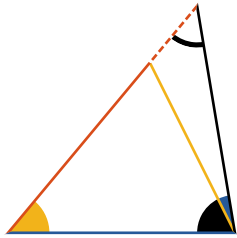
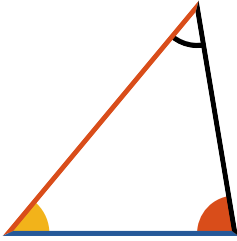
for if ▲ < ▲

then — < — (pr. I.24),

which is also contrary to the hypothesis:

∴ ▲ > ▲.

Q. E. D.



Book I. Prop. XXVI. Theor. *If two triangles have two angles of the one respectively equal to two angles of the other ($\triangle = \triangle$ and $\triangle = \triangle$), and a side of the one equal to a side of the other similarly placed with respect to the equal angles, the remaining sides and angles are respectively equal to one another. Case I.*

Let --- and --- which lie between the equal angles be equal, then

$$\text{---} = \text{---}.$$

For if it be possible, let one of them --- be greater; make $\text{---} = \text{---}$, draw --- .

In $\triangle$ and $\triangle$ we have

$$\text{---} = \text{---}, \triangle = \triangle, \\ \text{---} = \text{---};$$

$$\therefore \triangle = \triangle \text{ (pr. I.4)}$$

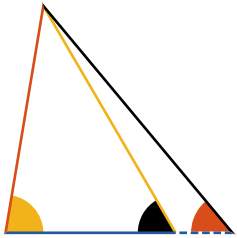
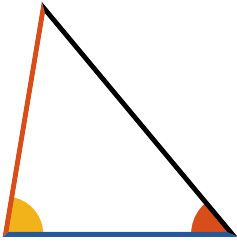
$$\text{but } \triangle = \triangle \text{ (hyp.)}$$

and therefore $\triangle = \triangle$, which is absurd;

hence neither of the sides --- and --- is greater than the other;

and $\therefore$ they are equal;

$$\therefore \text{---} = \text{---}, \text{ and } \triangle = \triangle \text{ (pr. I.4).}$$



Book I. Prop. XXVI. Theor. (cont.)
Case II.

Again, let  = , which lie opposite the equal angles  and .

If it be possible, let  > , then take  = , draw .

Then in  and  we have  = ,  =  and  = ,

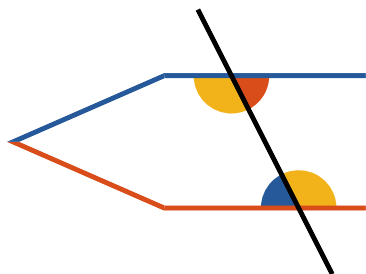
$\therefore$  =  (pr. I.4)

but  =  (hyp.)

$\therefore$  =  which is absurd (pr. I.16).

Consequently, neither of the sides  or  is greater than the other, hence they must be equal. It follows (pr. I.4) that the triangles are equal in all respects.

Q. E. D.

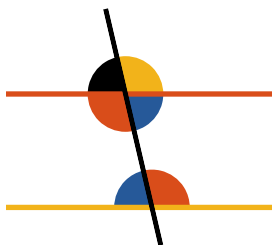


Book I. Prop. XXVII. Theor. *If a straight line (—) meeting two other straight lines (— and —) makes with them the alternate angles (▲ and ▼; ◐ and ◑) equal, these two straight lines are parallel. If*

— be not parallel to — they shall meet when produced.

If it be possible, let those lines be not parallel, but meet when produced; then the external angle ▼ is greater than ▲ (pr. I.16), but they are also equal (hyp.), which is absurd: in the same manner it may be shown that they cannot meet on the other side; ∴ they are parallel.

Q. E. D.



Book I. Prop. XXVIII. Theor. *If a straight line (—), cutting two other straight lines (— and —), makes the external equal to the internal and opposite angle, at the same side of the cutting line (namely $\blacktriangle = \triangle$ or $\triangle = \triangle$), or if it makes the two internal angles at the same side ($\triangle$ and $\triangle$, or $\triangle$ and $\triangle$) together equal to two right angles, those two straight lines are parallel.*

First, if $\blacktriangle = \triangle$, then

$\blacktriangle = \triangle$ (pr. I.15),

$\therefore \triangle = \triangle$

$\therefore \text{—} \parallel \text{—}$ (pr. I.27).

Secondly, if $\triangle + \triangle = \text{semicircle}$,

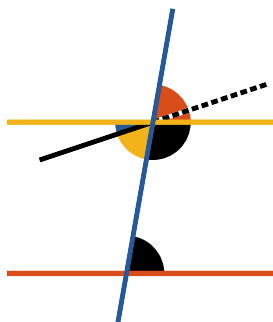
then $\triangle + \triangle = \text{semicircle}$ (pr. I.13),

$\therefore \triangle + \triangle = \triangle + \triangle$ (ax. III)

$\therefore \triangle = \triangle$

$\therefore \text{—} \parallel \text{—}$ (pr. I.27).

Q. E. D.



Book I. Prop. XXIX. Theor. *A*

straight line (—) falling on two parallel straight lines (— and —), makes the alternate angles equal to one another; and also the external equal to the internal and opposite angle on the same side; and the two internal angles on the same side together equal to two right angles. For

if the alternate angles  and  be not equal, draw —, making  =  (pr. I.23).

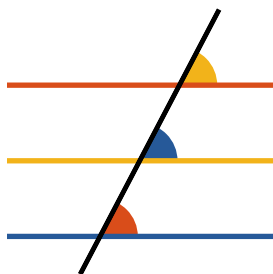
Therefore — — — — — || — (pr. I.27) and therefore two straight lines which intersect are parallel to the same straight line, which is impossible (ax. XII).

Hence  and  are not unequal, that is, they are equal:  =  (pr. I.15); $\therefore$  = , the external angle equal to the internal and opposite on the same side: if  be added to both, then  +  =

 =  (pr. I.13). That is to

say, the two internal angles at the same side of the cutting line are equal to two right angles.

Q. E. D.



Book I. Prop. XXX. Theor.
*Straight lines (— and —)
 which are parallel to the same straight
 line (—), are parallel to one
 another.*

Let — intersect { — };

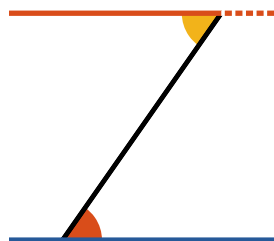
Then, $\triangle = \triangle = \triangle$ (pr. I.29),

$\therefore \triangle = \triangle$

$\therefore \text{ — } \parallel \text{ — }$ (pr. I.28).

Q. E. D.

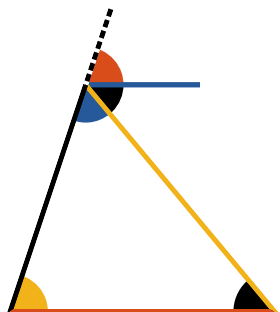
Book I. Prop. XXXI. Prob.



From a given point () to draw a straight line parallel to a given straight line ().

Draw  from the point  to any point  in ,
 make  =  (pr. I.23),
 then  $\parallel$  (pr. I.27).

Q. E. D.



Book I. Prop. XXXII. Theor. *If any side (—) of a triangle be produced, the external angle (◡) is equal to the sum of the two internal and opposite angles (◡ and ◡), and the three internal angles of any triangle taken together are equal to two right angles.*

Through the point  draw
 $\parallel$  (pr. I.31).

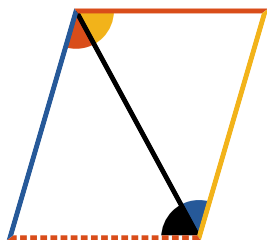
Then $\left\{ \begin{array}{l} \text{◡} = \text{◡} \\ \text{◡} = \text{◡} \end{array} \right\}$ (pr. I.29),

$\therefore \text{◡} + \text{◡} = \text{◡}$ (ax. II),

and therefore

$\text{◡} + \text{◡} + \text{◡}$
 $= \text{◡} = \text{◡} \quad (\text{pr. I.13}).$

Q. E. D.



Book I. Prop. XXXIII. Theor.
*Straight lines (——— and ———)
 which join the adjacent extremities of
 two equal and parallel straight lines
 (——— and), are themselves
 equal and parallel.*

Draw ——— the diagonal.

——— = (hyp.)

▲ = ▲ (pr. I.29)

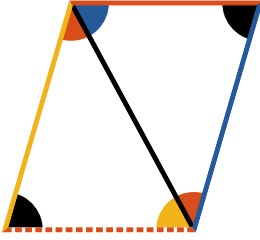
and ——— common to the two
 triangles;

∴ ——— = ———, and

▲ = ▲ (pr. I.4);

and ∴ ——— || ——— (pr. I.27).

Q. E. D.



Book I. Prop. XXXIV. Theor. *The opposite sides and angles of any parallelogram are equal, and the diagonal (—) divides it into two equal parts.*

$$\therefore \left\{ \begin{array}{l} \text{blue triangle} = \text{yellow triangle} \\ \text{red triangle} = \text{red triangle} \end{array} \right\} \text{ (pr. I.29)}$$

and — common to the two triangles.

$$\therefore \left\{ \begin{array}{l} \text{orange line} = \text{dashed red line} \\ \text{yellow line} = \text{blue line} \\ \text{black triangle} = \text{black triangle} \end{array} \right\} \text{ (pr. I.26)}$$

and  =  (ax. II).

Therefore the opposite sides and angles of the parallelogram are equal:

and as the triangles  and  are equal in every respect (pr. I.4), the diagonal divides the parallelogram into two equal parts.

Q. E. D.



Book I. Prop. XXXV. Theor. *Parallelograms on the same base, and between the same parallels, are (in area) equal.*

On account of the parallels,

$$\left. \begin{array}{l} \text{Red triangle} = \text{Blue triangle}; \\ \text{Black triangle} = \text{White triangle}; \end{array} \right\} \text{(pr. I.29, I.34).}$$

and $\text{Blue triangle} = \text{Red triangle}$

But $\text{Yellow triangle} = \text{Yellow triangle}$ (pr. I.8)

$$\therefore \text{Yellow triangle} - \text{Blue triangle} = \text{Yellow triangle} - \text{Red triangle},$$

$$\text{and } \text{Yellow triangle} - \text{Blue triangle} = \text{Yellow triangle} - \text{Red triangle};$$

$$\therefore \text{Yellow triangle} = \text{Yellow triangle}.$$

Q. E. D.



Book I. Prop. XXXVI. Theor. *Parallelograms ( and ) on equal bases, and between the same parallels, are equal.*

Draw  and 

 =  =

 (pr. I.34, hyp.);

$\therefore$  = and $\parallel$ ;

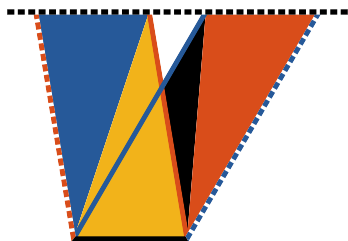
$\therefore$  = and $\parallel$  (pr. I.33)

and therefore  is a parallelogram;

but  =  =  (pr. I.35)

$\therefore$  =  (ax. I).

Q. E. D.



Book I. Prop. XXXVII. Theor. *Tri-*

angles  and  on the same base (—) and between the same parallels are equal.

Draw  ||  } (pr. I.31)
 ||  }

Produce

 and  are parallelograms on the same base and between the same parallels, and therefore equal (pr. I.35).

$\therefore \left\{ \begin{array}{l} \text{blue and yellow parallelogram} = \text{twice } \text{yellow triangle} \\ \text{black and orange parallelogram} = \text{twice } \text{black triangle} \end{array} \right\} \text{ (pr. I.34)}$

$\therefore \text{yellow triangle} = \text{black triangle}.$

Q. E. D.



Book I. Prop. XXXVIII. Theor.
*Triangles ( and ) on equal bases
and between the same parallels are
equal.*

Draw || } (pr. I.31)
and  ||  }

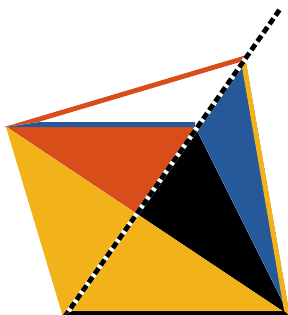
$$\text{Yellow triangle} = \text{Blue triangle} \text{ (pr. I.36);}$$

but = twice (pr. I.34),

and  = twice  (pr. I.34),

$$\therefore \text{orange triangle} = \text{blue triangle} \text{ (ax. VII).}$$

Q. E. D.



Book I. Prop. XXXIX. Theor.

Equal triangles  and  on the same base () and on the same side of it, are between the same parallels. If  , which joins the vertices of

the triangles, be not $\parallel$  , draw  $\parallel$  (pr. I.31), meeting .

Draw .

$\therefore$  $\parallel$  (const.)

 =  (pr. I.37);

but  =  (hyp.);

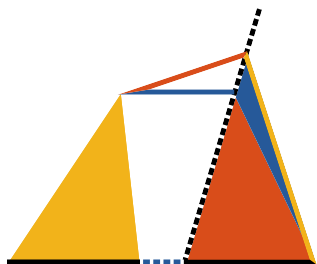
$\therefore$  = .

a part equal to the whole, which is absurd.

$\therefore$  $\nparallel$ ; and in the same manner it can be demonstrated that no other line except  is $\parallel$ .

$\therefore$  $\parallel$ .

Q. E. D.



Book I. Prop. XL. Theor. *Equal triangles (▲ and ▲) on equal bases, and on the same side, are between the same parallels. If ——— which joins the ver-*

tices of triangles be not || ———, draw ——— || ——— (pr. I.31), meeting

Draw ———.

∴ ——— || ——— (const.)

▲ = ▲ (pr. I.38)

but ▲ = ▲ (hyp.)

∴ ▲ = ▲,

a part equal to the whole, which is absurd.

∴ ——— ∦ ———: and in the same manner it can be demonstrated that no other line except ——— is || ———:

∴ ——— || ———.

Q. E. D.



Book I. Prop. XLI. Theor. *If a parallelogram  and a triangle  are upon the same base  and between the same parallels  and , the parallelogram is double the triangle.*

Draw  the diagonal.

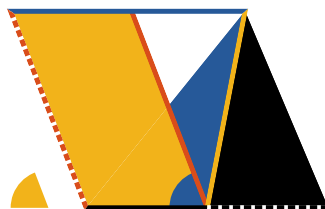
Then  =  (pr. I.37)

 = twice  (pr. I.34)

$\therefore$  = twice .

Q. E. D.

Book I. Prop. XLII. Prob.



To construct a parallelogram equal to
a given triangle  and having an
angle equal to a given rectilinear angle
().

Make  =  (pr. I.10)
Draw .

Make  =  (pr. I.23)

Draw   
   } (pr. I.31)

 = twice  (pr. I.41)

but  =  (pr. I.38)

$\therefore$  = .

Q. E. D.



Book I. Prop. XLIII. Theor. *The*

complements  *and*  *of the parallelograms which are about the diagonal of a parallelogram are equal.*



Q. E. D.

Book I. Prop. XLIV. Prob.



To a given straight line (—) to apply a parallelogram equal to a given triangle (▼), and having an angle equal to a given rectilinear angle (▲).

Make $\text{yellow parallelogram} = \text{red triangle}$ with $\text{blue triangle} = \text{yellow triangle}$ (pr. I.42)
and having one of its sides conterminous with and in continuation
of —.

Produce — till it meets — ||;
draw — produce it till it meets continued;
draw || meeting — produced and produce

$\text{yellow parallelogram} = \text{blue parallelogram}$ (pr. I.43)
but $\text{yellow parallelogram} = \text{red triangle}$ (const.)

$\therefore \text{blue parallelogram} = \text{red triangle}$;

and $\text{blue triangle} = \text{red triangle} = \text{black triangle} = \text{yellow triangle}$ (pr. I.29, const.).

Q. E. D.

Book I. Prop. XLV. Prob.



To construct a parallelogram equal to
a given rectilinear figure () and
having an angle equal to a given rec-
tilinear angle ().

Draw  and  dividing the rectilinear figure into triangles.

Construct  =  having  =  (pr. I.42)

to  apply  =  having  =  (pr. I.44)

to  apply  =  having  =  (pr. I.44)

$$\therefore \text{  =  }$$

and  is a parallelogram (pr. I.29, I.14, I.30)

having  =  .

Q. E. D.

Book I. Prop. XLVI. Prob.

Upon a given straight line (—) to
construct a square.



Draw — $\perp$ and = — (pr. I.11, I.3)

Draw — $\parallel$ —,
and meeting — drawn $\parallel$ —.



In

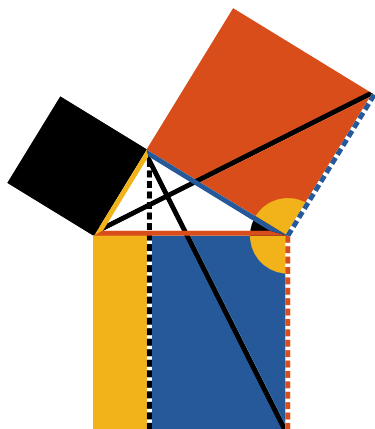
— = — (const.)

— = a right angle (const.)

$\therefore$ — = — = a right angle (pr. I.29),
and the remaining sides and angles must be equal (pr. I.34).

And $\therefore$ — is a square (def. 30).

Q. E. D.



Book I. Prop. XLVII. Theor. *In*

a right angled triangle  *the square on the hypotenuse*  *is equal to the sum of the squares of the sides* ( *and* ).

On , ,  describe squares (pr. I.46)

Draw  ||  (pr. I.31); also draw  and .

$$\text{[Diagram of a yellow sector]} = \text{[Diagram of a yellow sector]}.$$

To each add  : $\therefore$  = ,

$$\text{[Red line segment]} = \text{[Dashed red line segment]} \text{ and } \text{[Blue line segment]} = \text{[Dashed blue line segment]};$$

$$\therefore \text{[Blue triangle]} = \text{[Red triangle]} \quad (\text{pr. I.4}).$$

Again $\therefore$  || 

$$\text{[Red square]} = \text{twice } \text{[Red triangle]} \quad (\text{pr. I.41}),$$

$$\text{and } \text{[Blue square]} = \text{twice } \text{[Blue triangle]} \quad (\text{pr. I.41});$$

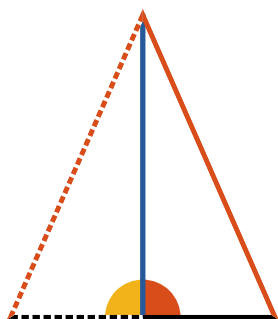
$$\therefore \text{[Red square]} = \text{[Blue square]}.$$

In the same manner it may be shown that

$$\text{[Black square]} = \text{[Yellow rectangle]};$$

$$\text{hence } \text{[Black square]} + \text{[Red square]} = \text{[Yellow rectangle]} + \text{[Blue rectangle]}.$$

Q. E. D.



Book I. Prop. XLVIII. Theor. *If the square of one side (—) of a triangle is equal to the squares of the other two sides (— and —), the angle (◡) subtended by that side is a right angle.*

Draw $\perp$ — and
 $=$ — (pr. I.11, I.3)
 and draw also.

Since = — (const.)
 $\text{.....}^2 = \text{---}^2$;

$$\therefore \text{.....}^2 + \text{---}^2 = \text{---}^2 + \text{---}^2$$

but 2 + — 2 =
 2 (pr. I.47),

and

$$\text{---}^2 + \text{---}^2 = \text{---}^2 \text{ (hyp.)}$$

$$\therefore \text{.....}^2 = \text{---}^2,$$

$$\therefore \text{.....} = \text{---};$$

and $\therefore$ ◡ = ◡ (pr. I.8),

consequently ◡ is a right angle.

Q. E. D.